Avik Solanki

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Education

University of Illinois Urbana-Champaign Champaign, IL
Bachelor of Science in Computer Engineering, Minor in Data Science *May 2029*

- **Relevant Coursework:** Computer Systems and Programming, Analog Signal Processing, Data Structures & Algorithms, Discrete Mathematics, Linear Algebra, Differential Equations, Integral Calculus, Physics: Mechanics, Physics: Electricity & Magnetism

Experience

Software and Robotics Engineer December 2025 – August 2026
DataraAI *Palo Alto, CA*

- Built a scalable Python and OpenCV ETL pipeline to ingest video data, extract frame-level datasets, and support high-volume computer vision preprocessing workloads.
- Implemented configurable frame-tagging logic to automate annotation, reduce manual labeling effort, and accelerate preprocessing and model-training iterations.
- Migrated local video and image datasets to Azure Blob Storage, improving reliability, enabling secure global access, and supporting distributed dataset management.
- Integrated Azure Cosmos DB to index pipeline metadata and processing status, using composite indexes to improve data organization and retrieval performance.
- Developed and tested robotic manipulation workflows in NVIDIA Isaac Sim, integrating computer vision, inverse kinematics, and physics-based simulation for autonomous connector insertion.

Projects

Vision-Guided Robotic Ethernet Insertion | *Isaac Sim, Python, OpenCV, PhysX* May 2026 – August 2026

- Developed an end-to-end RJ45 insertion system in NVIDIA Isaac Sim 6.0 using a Franka Panda, synchronized eye-in-hand RGB stereo cameras, YOLOE segmentation, Lula IK, and PhysX GPU simulation.
- Designed a perspective-correct port-center estimator that reconstructed the RGB front rim, calculated the diagonal intersection, and triangulated the 3D opening center without runtime ground truth or tuned XYZ offsets.
- Implemented a hybrid stereo visual-servo and kinematic handoff controller that locked a stable multi-frame world target before wrist-camera occlusion and completed the final approach through bounded Cartesian motion.
- Built a two-stage insertion state machine with ToolCenter feedback, quaternion orientation hold, and attachment checks, achieving 48/48 commands, 9.96 mm depth, 0.265 mm lateral drift, and 0.028° orientation error.

PrizePicker | *Python, Selenium, Flask, HTML, CSS* January 2026

- Engineered a Python and Selenium data-extraction pipeline to collect thousands of real-time player-prop data points across sportsbooks while handling dynamic DOM loading and anti-bot protections.
- Architected a fault-tolerant ingestion pipeline with automated session recovery, mitigating data loss from asynchronous DOM updates and maintaining near 100% player-prop capture across multiple markets.
- Developed a statistical arbitrage algorithm that cross-referenced FanDuel and PrizePicks lines to identify discrepancies and calculate expected value for positive-EV wagering opportunities.
- Built a responsive full-stack dashboard with Flask, HTML, and CSS to visualize collected data, using dynamic slip generation and directional indicators to streamline real-time decision-making.

Technical Skills

Languages: Python, C, Java, JavaScript, SQL, HTML, CSS, JSON
Cloud and Databases: Azure Blob Storage, Azure Cosmos DB, MongoDB
Robotics and Simulation: NVIDIA Isaac Sim 6.0, PhysX, Lula IK, YOLOE
Developer Tools: Git, GitHub, CI/CD, VS Code, Linux, Bash, Shell, Onshape